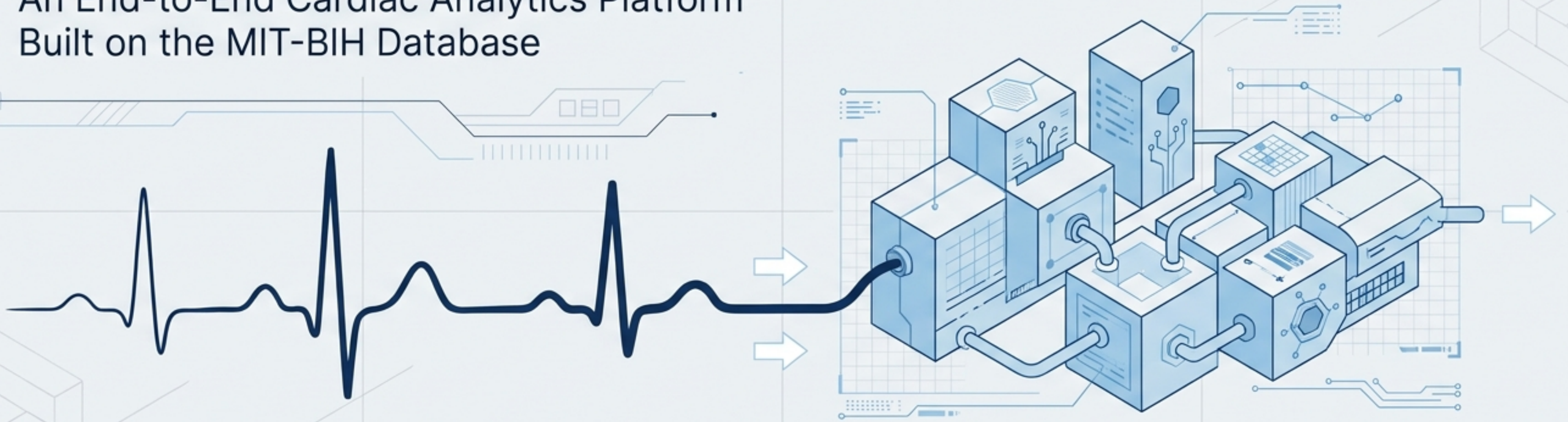


CardiAnalytics Pro: Bridging the Gap from AI Research to Clinical Reality

An End-to-End Cardiac Analytics Platform
Built on the MIT-BIH Database



Vishesh Gupta

Guided by Dr. Farzil Kidwai
MAIT Delhi, B.Tech CSE Final Year, 2026

The First Line of Defense Against 17.9 Million Annual Fatalities

THE STAKES

17.9 Million

Lives claimed annually by
Cardiovascular Disease
(WHO, 2023).



THE FOUNDATION



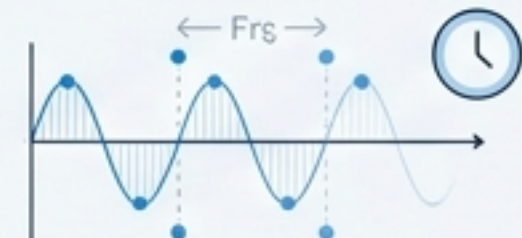
Sampled at real-time
360 Hz frequency



MIT-BIH Arrhythmia
1980: Developed by MIT
and Beth Israel Hospital



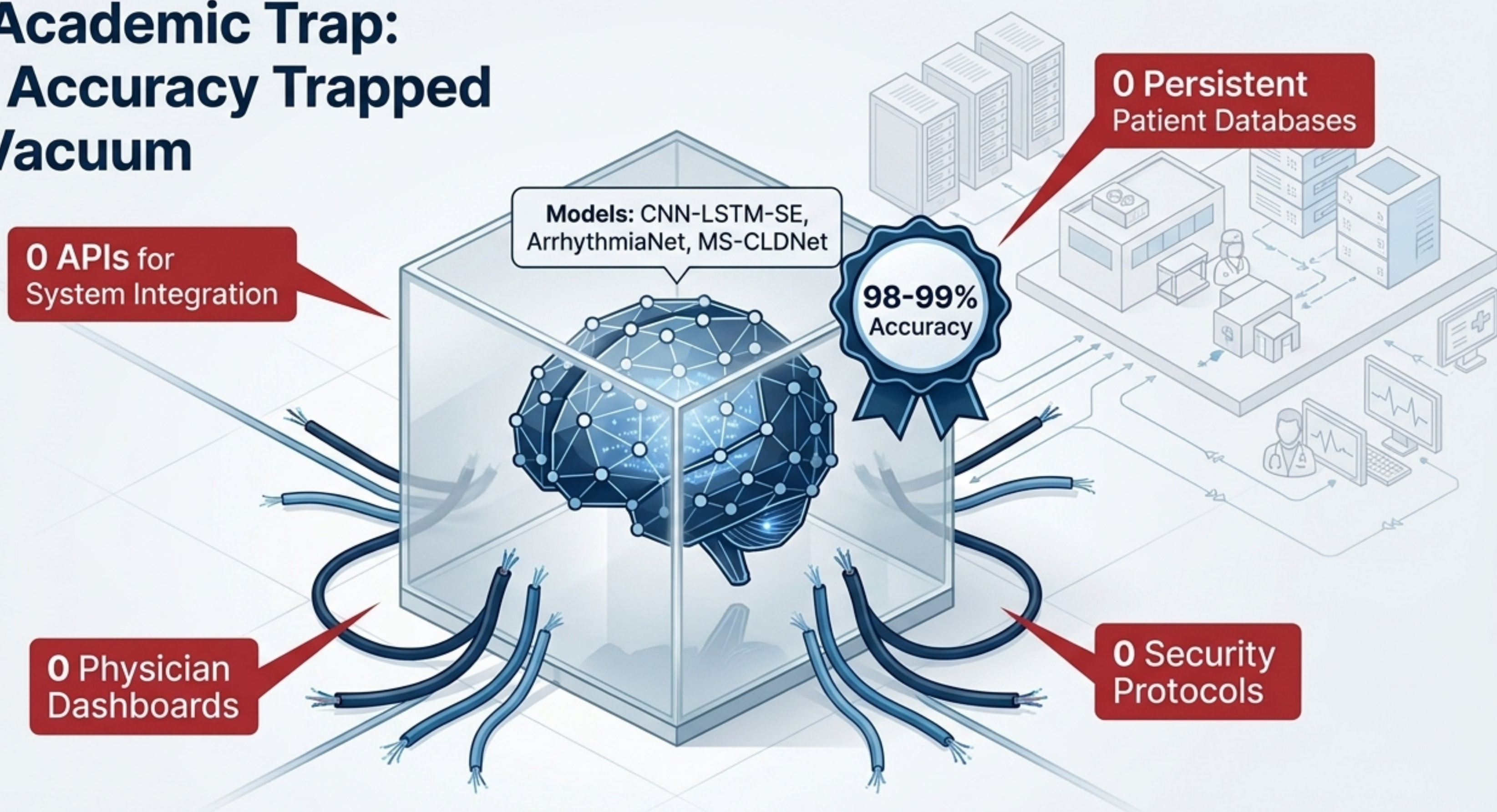
48 ambulatory ECG
recordings from
47 subjects



>3,000 published
studies cite as the
gold standard

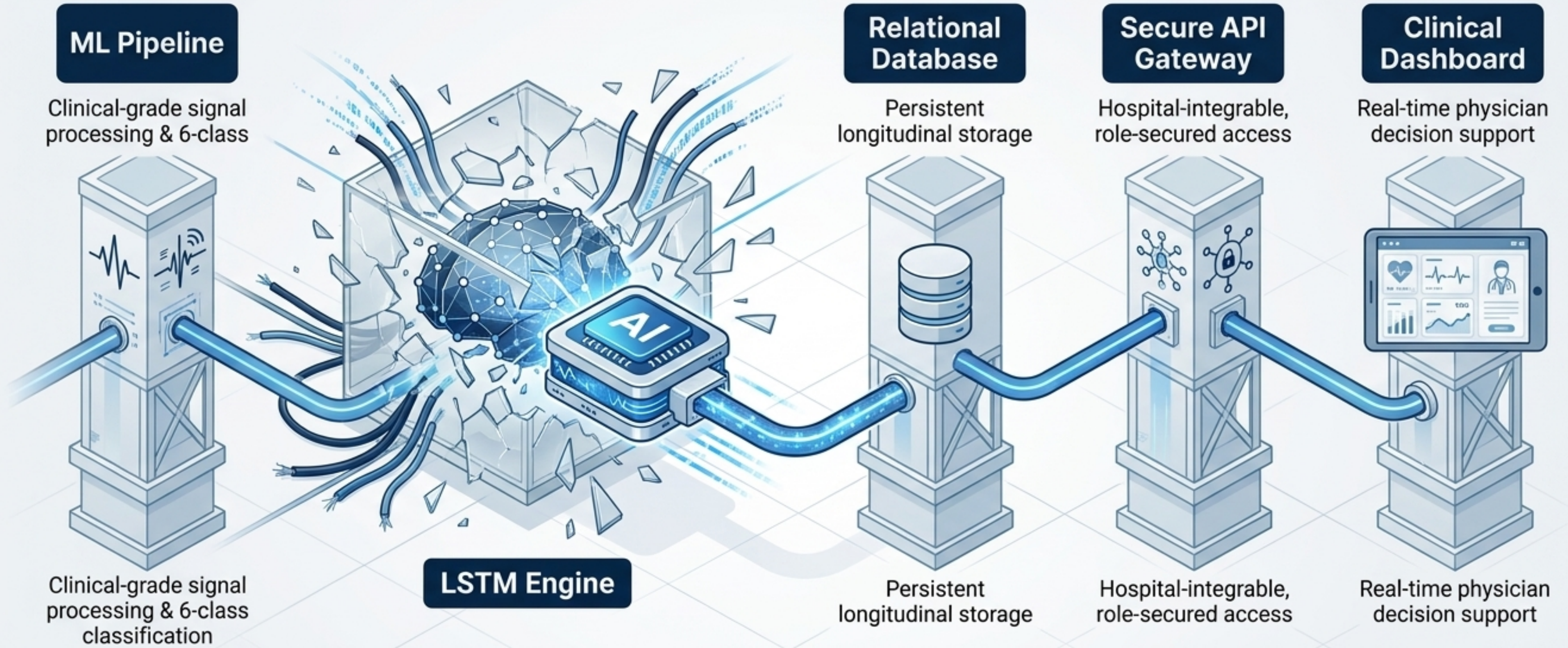


The Academic Trap: High Accuracy Trapped in a Vacuum

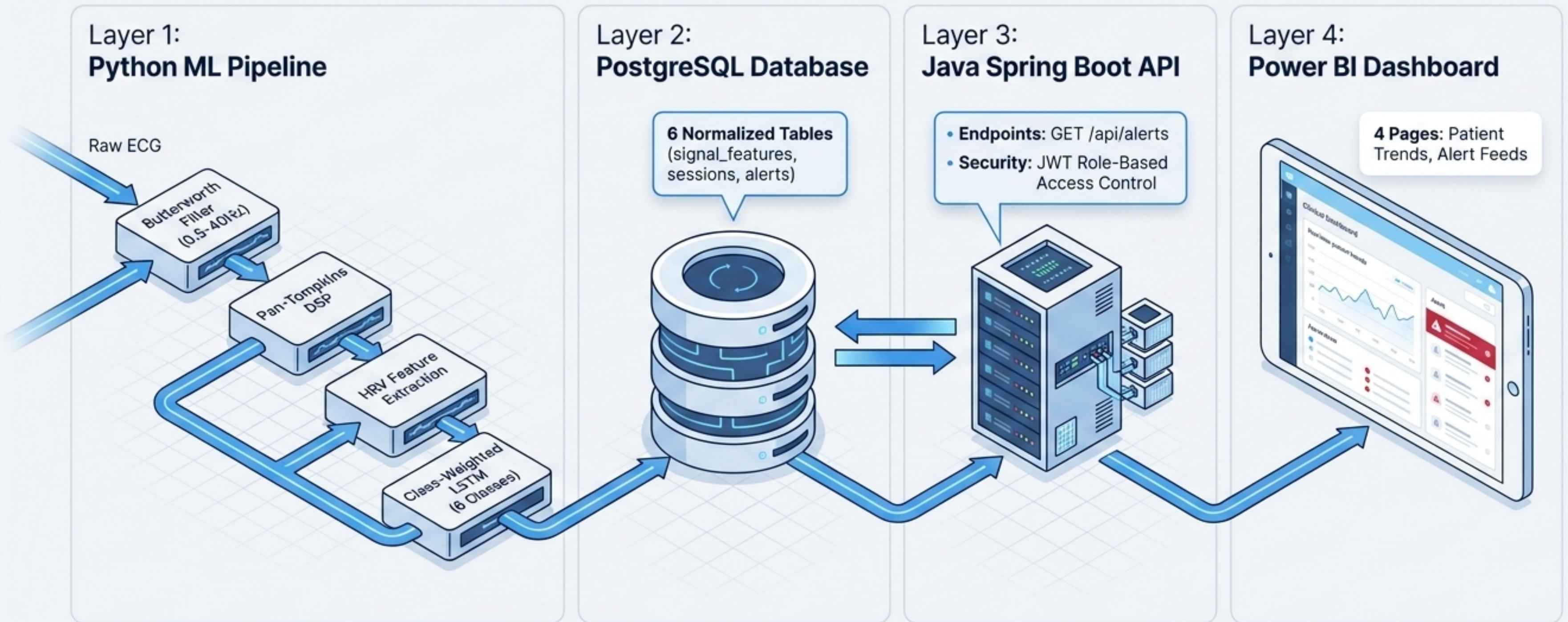


Key Takeaway: Recent studies produce non-deployable models. An isolated classifier cannot trigger a clinical intervention.

Moving from Beat Classification to Patient Management



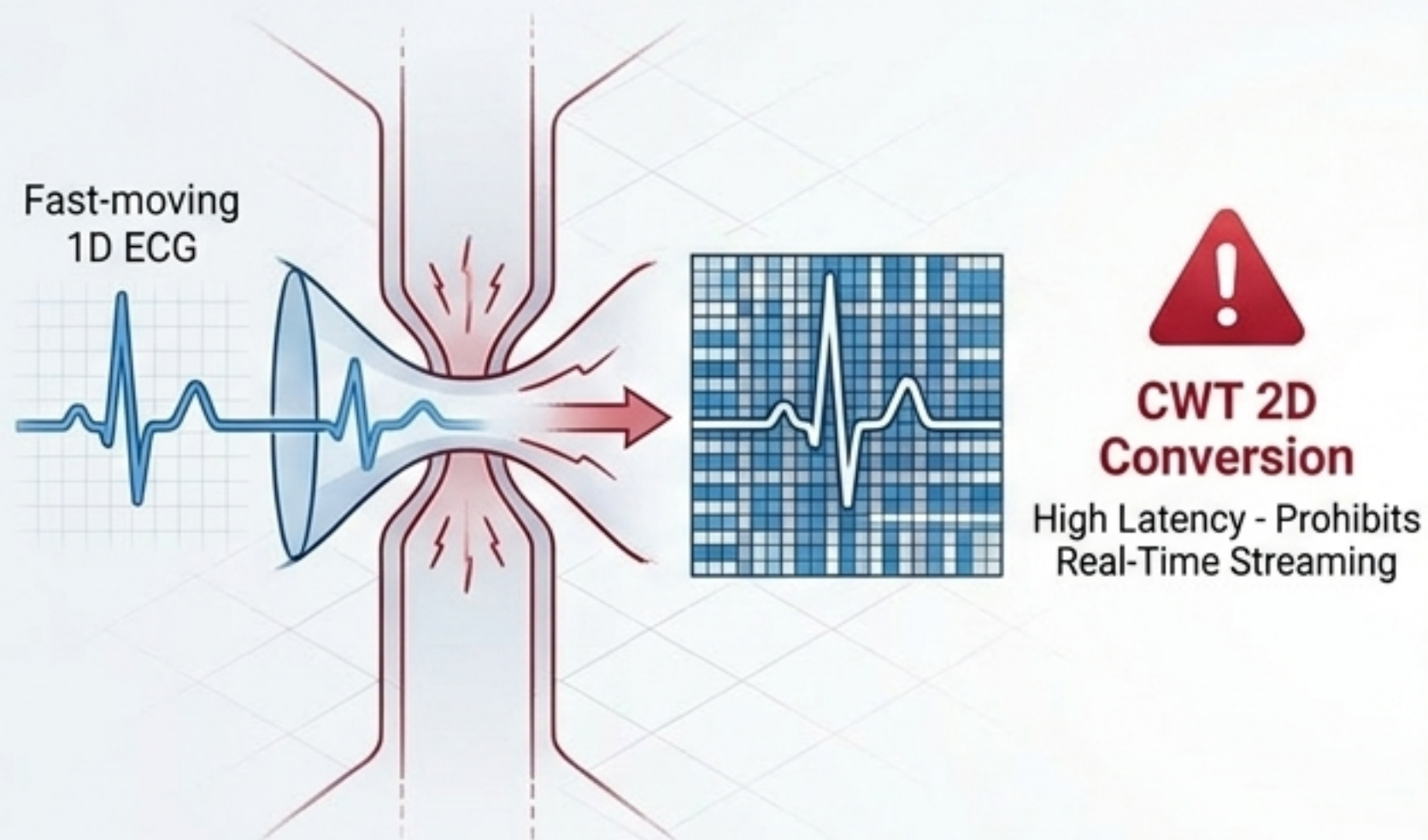
The CardiAnalytics Pro Master Architecture



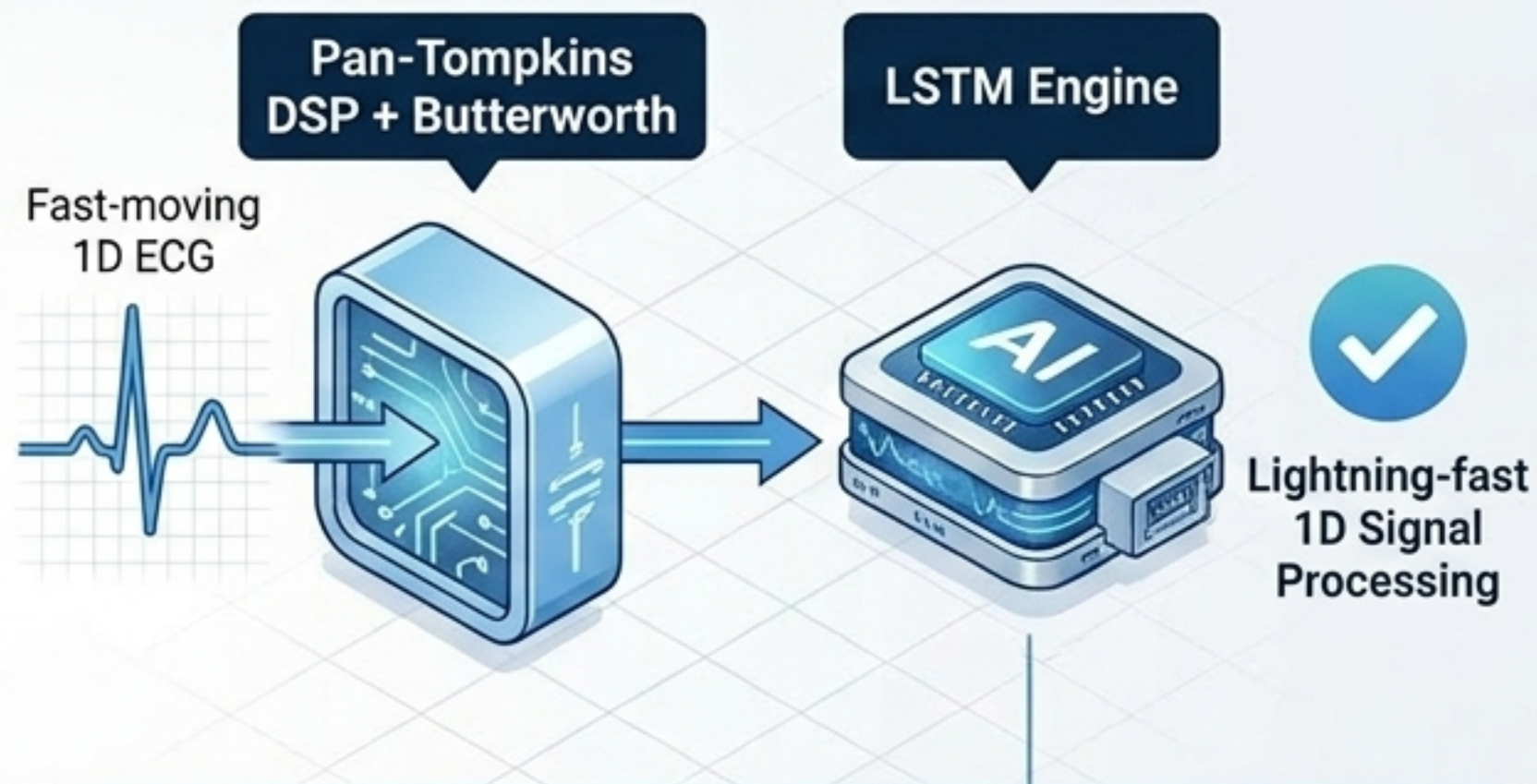
An end-to-end framework enabling a detected arrhythmia to become an actionable physician alert within milliseconds.

Layer 1: Clinical-Grade Signal Processing at 360 Hz

ArrhythmiaNet & Competitors

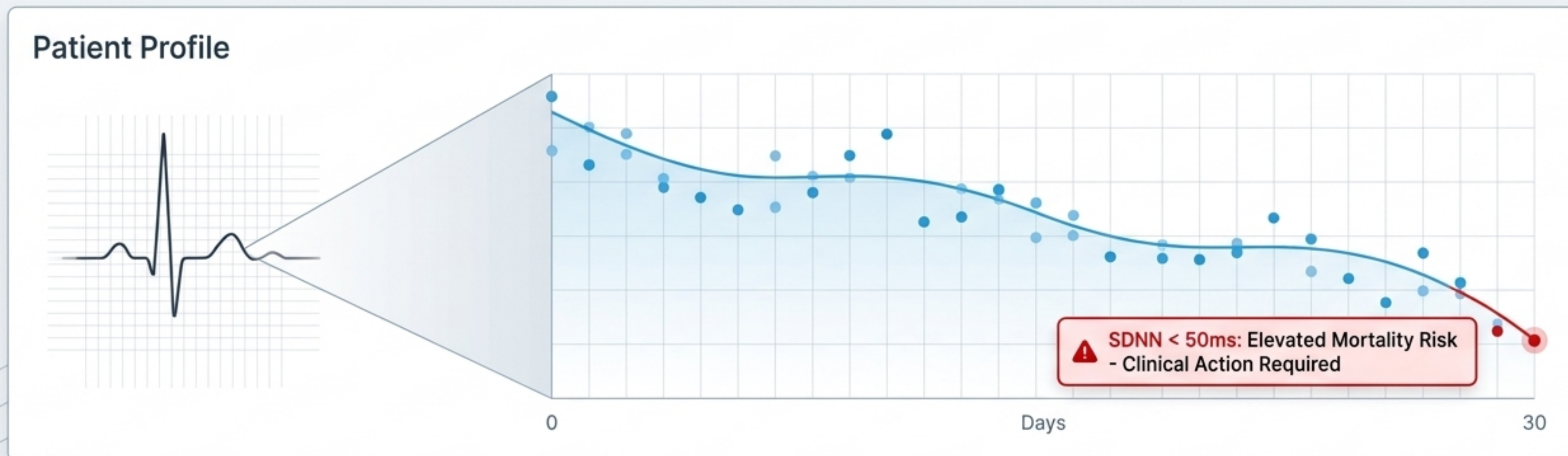


CardiAnalytics Pro



- **Pan-Tompkins:** 40-year clinical gold standard
- **Sensitivity:** >99% on real-world noisy signals
- **Capability:** Processes live ECG streams at 360 samples/second
- **Classifier:** Class-weighted LSTM (>90% accuracy, >0.85 macro F1 for 6 classes)

Layer 2: Beyond the Beat with Multi-Dimensional HRV Extraction



1
SDNN

Overall cardiac risk predictor

2
RMSSD

Short-term vagal tone / stress indicator
(Normal: 20-50ms)

3
pNN50

Parasympathetic nervous system health

4
Mean HR

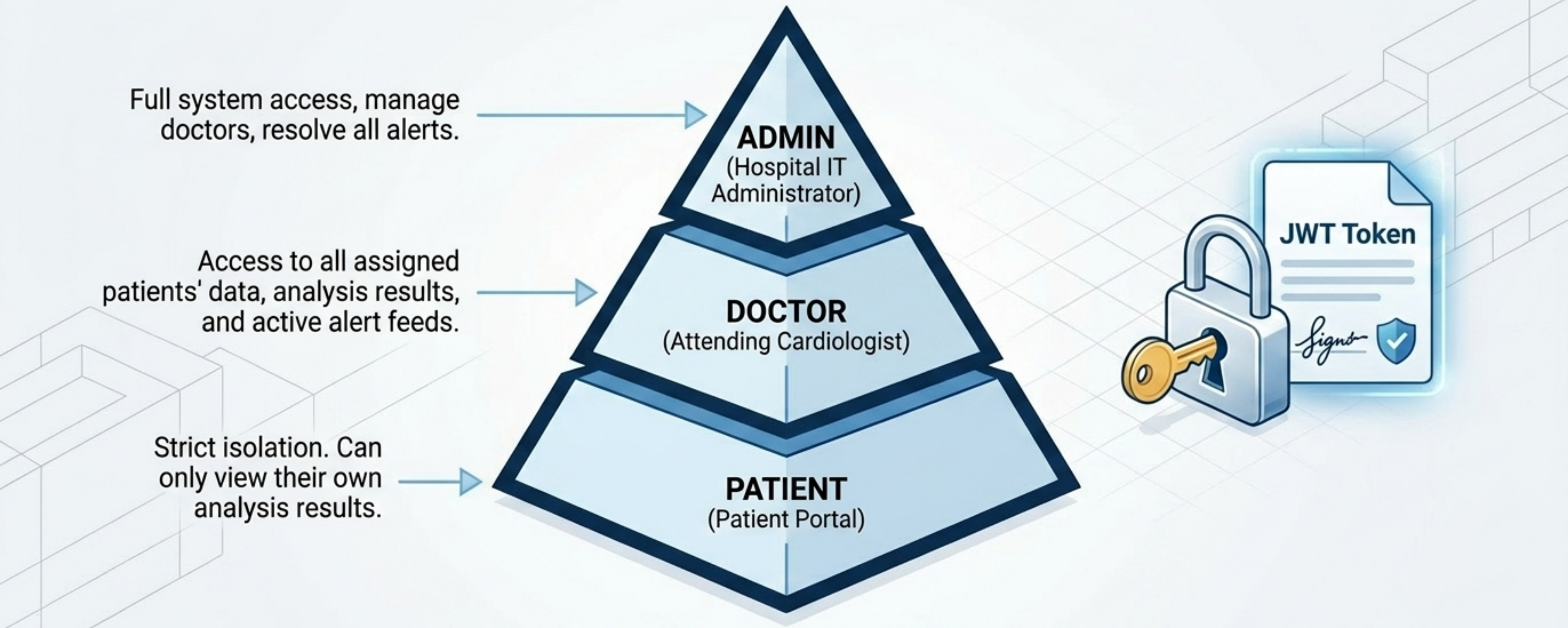
Baseline heart rate classification

5
SpO2 Estimate

Blood oxygen saturation via Lambert-Beer model

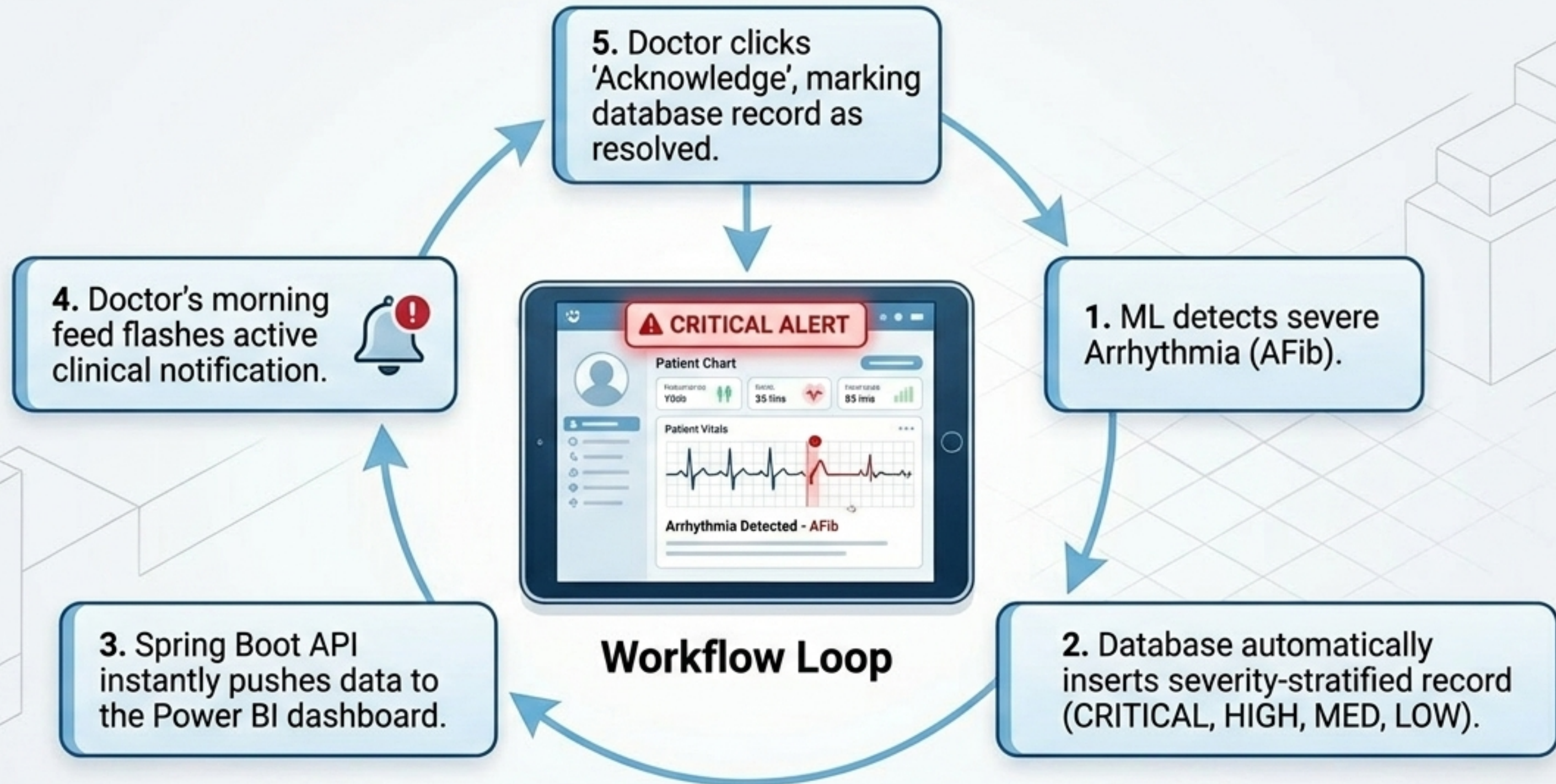
Competitors classify isolated beats. CardiAnalytics Pro persists HRV data to spot deteriorating trends weeks before a critical event.

Layer 3: HIPAA & DISHA Compliant Role-Based Architecture



A medical model without access control is an immediate legal liability. CardiAnalytics Pro is the only system reviewed built with native RBAC.

Layer 4: Closing the Loop with Real-Time Clinical Decision Support



Detection without notification is clinically worthless. This workflow transforms a passive algorithm into an active clinical partner.

System Synthesis: CardiAnalytics Pro vs. Recent Literature

Dimensions	Paper 1 (CNN-LSTM-SE)	Paper 2 (ArrhythmiaNet)	Paper 3 (Lightweight CNN)	Paper 4 (MS-CLDNet)	CardiAnalytics Pro
Dataset (MIT-BIH)	Yes	Yes	Yes	Yes	Yes
Bandpass Filtering	EEMD	CWT	Std	Wavelet	Butterworth 0.5-40Hz
HRV Features	No	No	No	No	5 Metrics
REST API	No	No	No	No	3 Endpoints
Database Storage	No	No	No	No	PostgreSQL 6 Tables
JWT Security/RBAC	No	No	No	No	3 Roles
Clinical Dashboard	No	No	No	No	Power BI
Alerting System	No	No	No	No	Severity Alerts

Real-World Performance Over Benchmark Vanity

Benchmark Vanity

99.8% Academic Accuracy



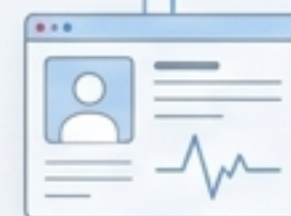
- Evaluated in sterile environments
- Utilizes slow 2D conversions
- Brittle, struggles with live streaming data

Clinical Efficacy

90%+ Target Accuracy
& **>0.85 Macro F1**

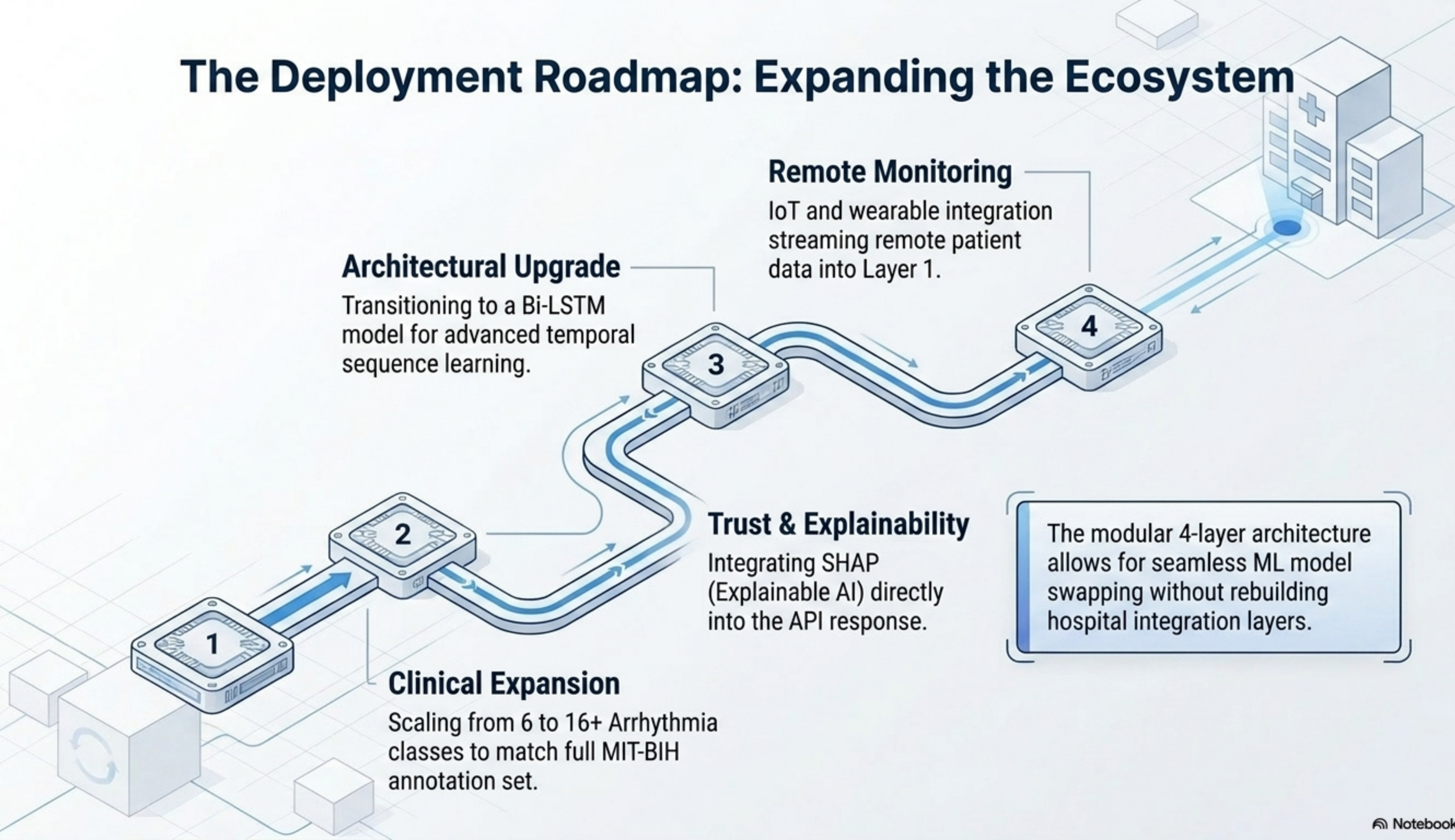


- Real-time capable at 360 Hz
- Resilient to imbalanced MIT-BIH data via class weights
- Clinically meaningful execution

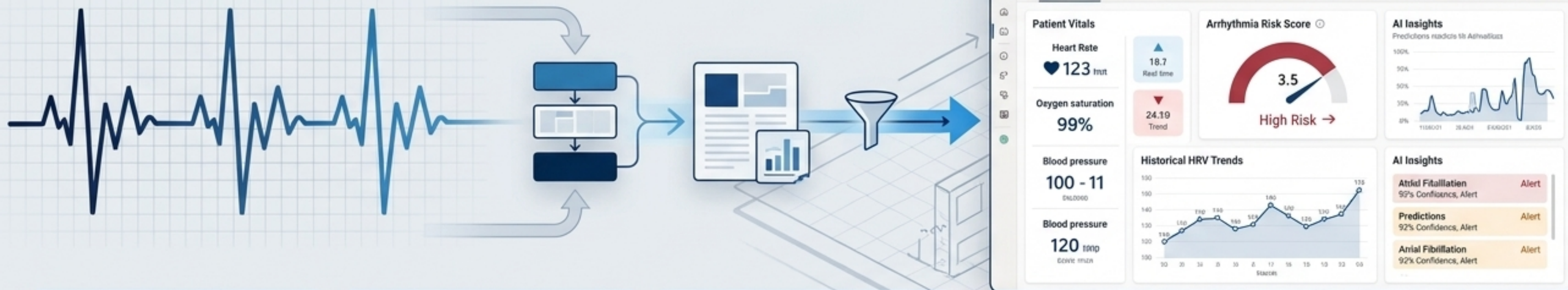


An accuracy of **99.8%** in a sterile Python notebook cannot save a life if it is computationally too heavy to run on a live hospital feed. We deliberately optimized for **deployability** and **macro F1-score** over raw benchmark accuracy.

The Deployment Roadmap: Expanding the Ecosystem



The Blueprint for Hospital Integration



Transformed isolated ML metrics into a 4-layer operational pipeline.



Pioneered multi-dimensional HRV tracking for longitudinal patient care.



Secured clinical data with enterprise-grade JWT role-based access.

CardiAnalytics Pro isn't just an algorithm; it is the definitive blueprint for how medical AI must be built, secured, and deployed.

Full
Research
Paper



GitHub
Repository

